

CSE 360 [Computer Interfacing]
Assignment 01

Total Marks: 25

1. What is key debouncing? Explain ways to avoid it with appropriate illustration of the mechanism. [3 marks]
2. Write down the mechanism of key identification of a Hex Keyboard. [4 marks]
3. Imagine you are trying to display your student ID using 7 segment displays. Illustrate the configuration of the 7 segment displays and explain the working mechanism using BCD to seven segment display configuration. [4 marks]
4. Describe how you can reduce the pin required to connect a LED Matrix display. [4 marks]
5. An 82C55 IC's Port A is connected to a pendrive and port B is connected to a display device. [10 marks]
 - a. Write down the control bits for the given connection of 82C55. Assume that pendrive will send data to the CPU.
 - b. Draw the handshaking diagram of port A only for the given scenario with appropriate pin numbers and signal labeling and explain the handshaking process.
 - c. Draw the handshaking diagram of port B only for the given scenario with appropriate pin numbers and signal labeling and explain the handshaking process.
 - d. Is there any pin left to connect a low level input device to the 82C55?
 - e. Describe the state of the 82C55 if the control bit is "10111010"